

Case Study - Azure

EDA analysis extract adls with Azure databricks and run queries of delta tables:

Since I am working with the Databricks Community Edition, which does not provide support for mounting Azure Data Lake Storage (ADLS), I will adapt my case study by simulating the ADLS component. Instead of mounting, I will upload the required datasets directly into the Databricks environment. This approach will allow me to proceed with the case study by performing exploratory data analysis (EDA) and executing Delta Lake queries, thereby closely replicating the intended workflow even without native ADLS integration.

Step 1: Prepare your dataset (CSV file)

Step 2: Upload this CSV file to Databricks Community Edition

- Login to community.cloud.databricks.com
- Click Data tab on left sidebar
- Click Add Data → Upload File
- Select your data.csv and upload

Step 3: Create a new notebook and start a cluster

- Go to Workspace > your user folder
- Click Create > Notebook
- Name it EDA_Case_Study
- Choose language Python
- Attach to a running cluster or start one

Step 4: Read the CSV file in your notebook

```
df = spark.read.format("csv") \
    .option("header", "true") \
    .option("inferSchema", "true") \
    .load("/Volumes/workspace/default/vol1/LoanData (1).csv")
df.show(5)
```

Step 5: Print the Schema of the table.

```
df.printSchema()
```

Step 6: Convert this Data Frame to a Delta Table

```
df.write.format("delta").mode("overwrite").saveAsTable("data_delta")
df.show()
```

Step 7: Run SQL queries on Delta table for EDA

1. Show first 10 records

```
SELECT * FROM data_delta LIMIT 2;
```

2. Count total records

```
SELECT COUNT(*) AS total_records FROM data_delta;
```

3. Calculating Average of applicant income and Average Loan Amount

```
SELECT
    AVG(CAST(ApplicantIncome AS INT)) AS AVG_Applicant_Income,
    AVG(CAST(LoanAmount AS INT)) AS AVG_Loan_Amount
FROM data_delta;
```

4. Average income by Education

```
SELECT
    Education,
    AVG(ApplicantIncome) AS avg_income,
    AVG(CoapplicantIncome) AS avg_coapp_income
FROM data_delta
GROUP BY Education;
```